

Exact off-diagonal telescope reassembly and critical phase-root boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-072 proved the matched same-shell phase-kernel leakage estimate and left three explicit off-diagonal shell families. This note determines exactly where those families go. They do not vanish and none may be discarded. Together with the R-071 linear term they reconstruct the R-069 mixed and control-current telescope exactly once.

The reconstruction yields a second exact fact. The apparent terminal phase-kernel leakage cancels on every rank stratum after restoring both the old-frame mixed first variation and the R-071 linear first variation. The result is a projector-free terminal current square. This is an algebraic completion, not a stochastic lower bound: after Wick subtraction it is the same adapted terminal coercivity problem isolated by R-069 and R-070.

The other possible route is audited as well. Within the declared termwise absolute largest-shell architecture at the raw R-050 root order, the two value-high branches are exactly critical. One branch requires $\theta > 1/4$ for shell decay and $\theta < 1/4$ for probability slack; the other requires simultaneously $\theta > 1/2$ and $\theta < 1/2$. The endpoint has neither decay nor a finite random Young exponent. This is a no-go only for that narrowly specified proof architecture. It is not a counterexample to A13 and not a no-go for signed cancellation, multilinear null structure, or an adapted root with positive causal gain.

All statements use the hash-pinned A1 production coefficients, the R-066 strict sharp-cube filtration, a fixed positive density floor, whole-shell predictable regular controls, and the six-real convention. The finite-low boundary is retained. Arbitrary finite-energy controls, controlled-shell one-use, the $q = 10/9$ Nelson theorem, floor removal, and infinite volume are not proved here.

2 Exact shell notation and the finite-low convention

Fix a high block $j_0 \leq j \leq J$ and start at

$$z_0 = U_J + A_{<j_0}, \quad z_{<j} = z_0 + \sum_{j_0 \leq k < j} a_k. \quad (2.1)$$

Suppress the production generator and spatial-direction sums and write

$$M_j = M(z_{<j}), \quad M_{j+1} = M(z_{<j} + a_j), \quad D_j = DM_j[a_j], \quad E_j = M_{j+1} - M_j - D_j. \quad (2.2)$$

Put

$$A = \sum_j a_j, \quad C = \sum_\ell b_\ell, \quad M_T = M(z_0 + A), \quad (2.3)$$

and, for $k < j$,

$$F_{k,j} = [DM(z_{\leq k}) - DM(z_{<k})][a_j]. \quad (2.4)$$

The two elementary telescopes are

$$\sum_{k < j} F_{k,j} = (DM_j - DM_0)[a_j], \quad (2.5)$$

$$E_{\text{tot}} := M_T - M_0 - DM_0[A] = \sum_j E_j + \sum_{k < j} F_{k,j}. \quad (2.6)$$

Let

$$w_j = M_j^T G, \quad \Delta w = (M_T - M_0)^T G, \quad C_j = \sum_{\ell < j} b_\ell, \quad c_j = M_j^T C_j, \quad c_T = M_T^T C. \quad (2.7)$$

If a low control current is present, C_j in (2.7) begins with $C_{<j_0}$ and every telescope below has its corresponding nonzero initial boundary. It is never reset to zero by notation.

Define the matched R-072 diagonal

$$\mathcal{D} = \sum_j \langle w_j, QE_j^T b_j \rangle, \quad (2.8)$$

the R-071 linear term

$$L = \langle w_0, QDM_0[A]^T C \rangle, \quad (2.9)$$

and the three off-diagonal families

$$\begin{aligned} \mathcal{O}_1 &= \sum_j \langle w_0 - w_j, QE_j^T b_j \rangle, \\ \mathcal{O}_2 &= \sum_j \sum_{\ell \neq j} \langle w_0, QE_j^T b_\ell \rangle, \\ \mathcal{O}_3 &= \sum_{k < j} \sum_{\ell} \langle w_0, QF_{k,j}^T b_\ell \rangle, \end{aligned} \quad (2.10)$$

with $\mathcal{O} = \mathcal{O}_1 + \mathcal{O}_2 + \mathcal{O}_3$.

3 Familywise off-diagonal reassembly

Each family has one precise role. First,

$$\mathcal{D} + \mathcal{O}_1 = \sum_j \langle w_0, QE_j^T b_j \rangle. \quad (3.1)$$

Thus $\mathcal{O}_1$ transports the nonlinear remainder root from w_j to w_0 . Second,

$$\mathcal{D} + \mathcal{O}_1 + \mathcal{O}_2 = \sum_{j,\ell} \langle w_0, QE_j^T b_\ell \rangle. \quad (3.2)$$

Thus $\mathcal{O}_2$ supplies all mismatched E_j - b_ℓ pairs. Finally, (2.5) gives

$$L + \mathcal{O}_3 = \sum_{j,\ell} \langle w_0, QDM_j[a_j]^T b_\ell \rangle. \quad (3.3)$$

In particular, $\mathcal{O}_3$ does not cancel L . It transports the base derivative DM_0 to the local derivative DM_j .

Adding (3.1)–(3.3) and using (2.6) proves the exact endpoint identity

$$\boxed{L + \mathcal{D} + \mathcal{O} = \langle w_0, Q(M_T - M_0)^T C \rangle.} \quad (3.4)$$

No inequality, conditioning, or rank assumption is used. In particular, the independent R-072 fixture with $|\mathcal{O}|/|\mathcal{D}| > 4087$ is not an anomaly: the full endpoint transport is stored predominantly in the off-diagonal families in that fixture.

4 Return to the R-069 current telescope

Define the separated base first variations

$$S_C = \langle w_0, QM_0^T C \rangle, \quad S_G = \langle w_0, QDM_0[A]^T G \rangle, \quad (4.1)$$

and

$$N_G = \langle w_0, QE_{\text{tot}}^T G \rangle. \quad (4.2)$$

For an empty initial high-control current, the R-069 polarizations give

$$\sum_j \mathcal{R}_j^\times = \langle c_T, Qw_T \rangle, \quad \sum_j \mathcal{R}_j^C = \frac{1}{2}|c_T|_Q^2, \quad (4.3)$$

where $w_T = w_0 + \Delta w$. The raw Gaussian channel gives

$$\sum_j \mathcal{R}_j^{G,\text{raw}} = S_G + N_G + \frac{1}{2}|\Delta w|_Q^2. \quad (4.4)$$

Equations (3.4) and (4.3) imply

$$\boxed{\mathcal{D} + \mathcal{O} + \frac{1}{2}|\Delta w + c_T|_Q^2 = \sum_j (\mathcal{R}_j^\times + \mathcal{R}_j^C) - S_C - L + \frac{1}{2}|\Delta w|_Q^2.} \quad (4.5)$$

Adding (4.4) yields the decisive full identity

$$\boxed{L + N_G + \mathcal{D} + \mathcal{O} + \frac{1}{2}|\Delta w + c_T|_Q^2 = \sum_j (\mathcal{R}_j^{G,\text{raw}} + \mathcal{R}_j^\times + \mathcal{R}_j^C) - S_G - S_C.} \quad (4.6)$$

At the stochastic level, raw-to-Wick conversion restores the exact R-066 trace channel. The finite-low version retains

$$\mathbb{E} \sum_{j=j_0}^J (\mathcal{R}_j^{G,\text{Wick}} + \mathcal{R}_j^\times + \mathcal{R}_j^C) = \mathbb{E}(\mathcal{W}_{J+1} - \mathcal{W}_{j_0}). \quad (4.7)$$

Consequently, (4.6) is an exact equivalence back to the terminal translated current coercivity isolated in R-070. It is not a new lower bound. Dropping $\mathcal{W}_{j_0}$ or citing the desired Nelson moment to bound $\mathcal{W}_{J+1}$ would be respectively false or circular.

5 Projector-free completion and the phase kernel

It is useful to see precisely why the terminal square appeared blind in R-071. Put

$$\mathfrak{P}_{\text{mix}}(C) = \langle w_0, QE_{\text{tot}}^T C \rangle + \langle \Delta w, Qc_T \rangle + \frac{1}{2}|c_T|_Q^2. \quad (5.1)$$

If $B_x = \sum_r M_r(x)QM_r(x)^T$, then the same tensor is

$$\mathfrak{P}_{\text{mix}}(C) = C^T \{ [B_T - B_0]G - DM_0[A]QM_0^T G \} + \frac{1}{2}C^T B_T C. \quad (5.2)$$

The terms $M_T Q \Delta w$ and B_T in (5.2) are visible to the terminal analysis map $T_T C = (M_{r,T}^T C)_r$. After S_C and L have been split away, the square-invisible quotient is

$$(P_{\ker T_T} C) \cdot \sum_r E_{r,\text{tot}} Qw_{r,0}. \quad (5.3)$$

It is irreducible to the terminal square alone, but it is not irreducible in the full signed energy.

Indeed, ordinary algebra gives on every rank stratum

$$\boxed{S_C + L + \mathfrak{P}_{\text{mix}}(C) = \langle w_T, Qc_T \rangle + \frac{1}{2}|c_T|_Q^2.} \quad (5.4)$$

Therefore

$$\boxed{\frac{1}{2}|w_T|_Q^2 + S_C + L + \mathfrak{P}_{\text{mix}}(C) = \frac{1}{2}|w_T + c_T|_Q^2.} \quad (5.5)$$

No projector or pseudoinverse occurs. In particular, if $C \in \ker T_T$, then

$$S_C + L + \langle w_0, QE_{\text{tot}}^T C \rangle = 0. \quad (5.6)$$

The apparent kernel obstruction is created by separating terms which the full Taylor identity $M_0 + DM_0[A] + E_{\text{tot}} = M_T$ couples exactly.

For completeness, write a terminal value as $x = z + A = (u, s)$ and set

$$n_D = J_D x, \quad n_S = J_S x. \quad (5.7)$$

The exact Euclidean kernel projector is

$$P_{\ker T_x} = \begin{cases} \frac{n_D n_D^T}{|u|^2} + \frac{n_S n_S^T}{|s|^2}, & u \neq 0, \ s \neq 0, \\ \frac{n_D n_D^T}{|u|^2} + P_S, & u \neq 0, \ s = 0, \\ I_6, & u = 0. \end{cases} \quad (5.8)$$

Its ranks are 2, 3, and 6, so a smooth fixed-rank inverse cannot replace (5.5).

There is also an explicit formula for the nonlinear vector in (5.3). Write

$$M_r = [p_r, v_r], \quad \ell_r = Dq_r(z)[A], \quad \epsilon_r = q_r(x) - q_r(z) - \ell_r, \quad (5.9)$$

$$\beta_r = e_2^T Qw_{r,0}, \quad \mathcal{L}_\beta = \sum_r \beta_r \ell_r, \quad \mathcal{E}_\beta = \sum_r \beta_r \epsilon_r. \quad (5.10)$$

Since p_r is linear, the following identities are exact:

$$E_r^p = 0, \quad E_r^v = -2(\epsilon_r x + \ell_r A), \quad k := \sum_r E_r Qw_{r,0} = -2(\mathcal{E}_\beta x + \mathcal{L}_\beta A). \quad (5.11)$$

On the generic stratum this gives

$$(P_{\ker T_x} C) \cdot k = -2\mathcal{L}_\beta \left\{ \frac{(C \cdot n_D)(A \cdot n_D)}{|u|^2} + \frac{(C \cdot n_S)(A \cdot n_S)}{|s|^2} \right\}. \quad (5.12)$$

At $s = 0$, the second summand in braces is replaced by $(P_S C) \cdot (P_S A)$. At $u = 0$, the whole target is the kernel and one uses (5.11) directly. These formulas explain the rank jumps, while (5.4)–(5.5) show that no division by a small field is needed in the complete energy.

For either genuine phase line, R-072 differentiation gives

$$n_\alpha \cdot k = - \sum_r \langle DM_r(z)[A]^T J_\alpha A, Qw_{r,0} \rangle = -2\mathcal{L}_\beta(n_\alpha \cdot A). \quad (5.13)$$

This identity says nothing about the extra singlet radial kernel at $s = 0$, or the full kernel at $u = 0$; (5.5) covers those strata instead.

The R-071 production fixture is decisive. At the pinned floor the three slopes, all recomputed by the executable, are

$$N' = 0.0674999999999706, \quad L' = 0.0108750000000198, \quad S'_C = -0.0783749999999904. \quad (5.14)$$

Thus the separated linear term reinforces the nonlinear leakage. Only the restored base first variation gives $S'_C + L' + N' = 0$. Moreover terminal phase directions satisfy $d\rho_x[n_\alpha] = dm_{r,x}[n_\alpha] = 0$, hence the terminal Cartan curvature and Jacobi square do not cancel N' .

6 Scoped largest-shell absolute-route audit

Let

$$x_j = \|h_j\|_{\mathcal{H}_j}^2, \quad y_j = \|a_j\|_6^6, \quad N_j = 2^j. \quad (6.1)$$

The declared raw estimates are

$$\|b_j\|_2 \lesssim N_j^{-1} x_j^{1/2}, \quad (6.2)$$

$$\|a_j\|_6 \lesssim N_j^{-\theta} x_j^{\theta/2} y_j^{(1-\theta)/6}, \quad 0 \leq \theta \leq 1. \quad (6.3)$$

After expanding a term and choosing its largest control index r , exact strict-past causalization gives a normalized Gaussian factor

$$\|G_r\|_6 \lesssim N_r^{1/2} \mathfrak{g}_r, \quad (6.4)$$

with every fixed finite moment. The base-current difference family has an extra $N_j^{-1/2}$ and closes with budgets $X^{1/2}Y^{1/3}$ and a sixth moment. Comparable-index and derivative-high parts of the other two families also retain negative dyadic powers. Two value-high orientations are different.

For $\mathcal{O}_2$ with $j \gg \ell$, (6.2)–(6.4) give the dyadic weight

$$N_j^{1/2-2\theta} N_\ell^{-1}. \quad (6.5)$$

Decay in the largest shell requires $\theta > 1/4$. The total deterministic budget exponent is

$$\alpha_2 = \theta + \frac{1}{2} + \frac{1-\theta}{3} = \frac{5}{6} + \frac{2\theta}{3}. \quad (6.6)$$

A finite random Young exponent requires $\alpha_2 < 1$, or $\theta < 1/4$. At $\theta = 1/4$, (6.5) has no separation decay and $\alpha_2 = 1$.

For $\mathcal{O}_3$ with $j \gg k, \ell$, the weight is

$$N_j^{1/2-\theta} N_\ell^{-1}, \quad (6.7)$$

which requires $\theta > 1/2$. Its deterministic exponent is

$$\alpha_3 = \frac{1}{2} + \frac{\theta}{2} + \frac{1-\theta}{6} + \frac{1}{6} = \frac{5}{6} + \frac{\theta}{3}, \quad (6.8)$$

so random slack requires $\theta < 1/2$. Again the endpoint has zero decay and $\alpha_3 = 1$.

This is a formal obstruction to the stated termwise absolute two-budget Young architecture. At $\alpha = 1$, scaling both abstract positive budgets by a common factor leaves the unbounded random multiplier in the leading coefficient, while every additive random remainder is lower order in that scale. Moving θ to either side fixes one condition and violates the other. This argument does not exclude cancellations unavailable after absolute values, and it is not an integrated production counterexample.

A separated controlled finite-low factor produces the same endpoint warning: the schematic budget is $X^{1/2}Y^{1/2}$. It must remain in the signed finite-low telescope rather than be assigned a second absolute payment.

7 The positive-gain discriminator

Suppose a future adapted two-parameter root theorem improves the raw $H^{-1/2-\delta}$ order by γ derivatives. Put

$$\rho = \gamma - \delta, \quad \sigma = \frac{1}{2} - \rho. \quad (7.1)$$

The conflicts in (6.5)–(6.8) open precisely when $\rho > 0$. The limiting Young exponent is

$$p > \frac{3}{\rho} = \frac{3}{\gamma - \delta}, \quad (7.2)$$

where strict inequality allows the dyadic endpoint losses. Equivalently, the missing model has effective regularity $H^{-1/2+\rho}$, is tame under strict-past controls, retains every lower-chaos conversion, and has an $L^p(\Omega)$ norm for (7.2).

R-063's unshifted second balanced jet suggests an effective order near $H^{-0.3}$, corresponding formally to $\rho \simeq 0.2$ and ideal endpoint $p = 15$. R-063 explicitly does not prove the adapted, two-control, lower-chaos-complete operator-valued theorem required here. Substituting its unshifted estimate would therefore be an overclaim.

8 Evidence map, failed routes, and successor

The reusable proof map is now:

1. **Closed:** equations (3.1)–(3.4) classify every off-diagonal family. Each appears exactly once; none cancels locally.
2. **Closed:** equations (4.3)–(4.6) return the complete remainder to the R-069 raw current telescope, with the R-071 linear payment and both base first variations counted exactly once.
3. **Closed algebraically:** equations (5.4)–(5.5) give a projector-free terminal completion on the generic, singlet-tip, and doublet-collapse strata. For terminal-kernel controls, (5.6) is the exact signed cancellation.
4. **Failed route:** terminal square alone, terminal Cartan alone, or the already-separated R-071 linear term does not cancel the phase leakage. The exact production fixture (5.14) fixes the sign and attribution.
5. **Failed route:** the raw termwise absolute value-high $\mathcal{O}_2/\mathcal{O}_3$ Carleson architecture is endpoint-critical by (6.5)–(6.8). This is registered as NG-2026-07-24-A13-RAW-ABSOLUTE-OFFDIAGONAL-CARLESON.
6. **Still open:** converting the signed terminal completion into a cutoff-uniform stochastic lower bound. Returning to (4.7) reconstructs the already-open terminal coercivity and supplies no new sign.

The smallest honest successor has two possible forms. Either prove a signed strict-past terminal phase-root coercivity theorem which exploits (5.4) while retaining the range square and $-\mathcal{W}_{j_0}$, or prove the adapted positive-gain root described by (7.1)–(7.2) and use it only on the two value-high branches. Only then may localization and lower semicontinuity extend the result to every finite-energy drift, followed by controlled-shell one-use and Nelson synthesis.

9 Devil's-advocate review

1. **“Exact reassembly proves coercivity.” UPHELD as false.** Equation (4.7) shows that reassembly lands on the adapted terminal current itself. Bounding it with the desired Nelson moment would be circular.
2. **“The third off-diagonal family cancels the R-071 linear term.” UPHELD as false.** Equation (3.3) transports DM_0 to DM_j ; both contributions remain with the same endpoint Taylor increment.
3. **“The terminal square sees every derivative displacement.” UPHELD as false after the first variations are separated.** The rank-2/3/6 kernel in (5.8) and the nonzero fixture (5.14) refute that claim. The mitigation is the full signed completion (5.5), not a singular pseudoinverse.

4. **“The terminal Cartan or Jacobi term cancels the gauge leakage.” UPHELD as false.** A terminal phase displacement kills $d\rho$, every dm_r , and the terminal current, while (5.13)–(5.14) remain nonzero.
5. **“The endpoint choices $\theta = 1/4$ and $\theta = 1/2$ close the absolute estimate.” UPHELD as false.** They make the dyadic decay and the random Young slack vanish simultaneously.
6. **“The exponent conflict disproves A13.” DISMISSED.** It retires only the declared raw, termwise absolute, two-budget architecture. Signed telescope cancellation and a positive-gain adapted model remain viable.
7. **“R-063 already supplies the gain in (7.2).” UPHELD as false.** Its theorem is unshifted or deterministically shifted and explicitly excludes the adapted random-shift estimate needed here.
8. **“The high block has zero initial current by definition.” VALID with mitigation.** This is true only after absorbing the finite-low block into z_0 and retaining the corresponding $-\mathcal{W}_{j_0}$ boundary. No low term is deleted.
9. **“This result proves finite-energy one-use or Nelson.” UPHELD as an overclaim.** It closes the algebraic coupling and a scoped route audit only. The stochastic coercivity, finite-energy extension, controlled-shell one-use, and Nelson gates remain open.

10 Reproduction and independent audit

The primary executable derives the production Q_{Π} and floor from A1, checks one-, two-, and four-shell cases, verifies every familywise and full telescope identity, reconstructs the rank-stratified projectors, checks the explicit nonlinear vector, and recomputes the R-071 phase slopes. The non-importing executable rebuilds the Pauli frames and production coefficients, uses a different seed and two-to-four-shell fixtures, and independently checks the same completion and exponent boundary. The integrated verifier hash-pins R-069–R-072, the sources, note, PDF, and manifest before executing both children.

```
python codes/foundations/
  a13_classii_off_diagonal_telescope_critical_phase_root_reduction_verify.py
```

Result footer

Result ID: A13-CLASSII-OFF-DIAGONAL-TELESCOPE-CRITICAL-PHASE-ROOT-REDUCTION

Precise statement:	The R-072 off-diagonal families and R-071 linear term reconstruct R-069; both restored first variations give a projector-free terminal square.
Scope:	Fixed-floor A1/R-066; regular shell-predictable controls; finite-low boundary retained.
Dependencies:	A1; R-050; R-063; R-066; R-069--R-072.
Evidence grade:	ANALYTIC, EXACT, EXECUTED.
Reproduction command:	python codes/foundations/ a13_classii_off_diagonal_telescope_critical_phase_root_reduction_verify.py
Expected output:	35/35 + 27/27 + 51/51 = 113/113; exit 0.
Falsification gate:	A telescope, completion, kernel rank, phase, exponent, hash, or assertion gate fails.
Tier before / after:	T4 / T4; no promotion.
No-overclaim statement:	No stochastic lower bound, adapted gain, finite-energy one-use, Nelson, measure, removal, or T5--T7.
Next required action:	Prove adapted phase-root coercivity or a two-control gain $\rho > 0$, $p > 3/\rho$; then one-use and Nelson.